

Question 1. (30 points)

Consider an experiment with two factors, each with two levels.

Consider the two-factor linear model with interaction

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon, \quad \varepsilon \sim \text{i.i.d. } (0, \sigma^2).$$

Compare two designs with coded levels in $[-1, 1]$:

- **Design A (factorial):** full 2^2 factorial with coded levels $\{-1, +1\}$ for each factor, with r replicates per cell (total $n = 4r$).
- **Design B (random):** $n = 4r$ runs where $x_1, x_2 \stackrel{\text{i.i.d.}}{\sim} \text{Unif}[-1, 1]$, independent across runs.

Simulation part. Fix $(\beta_0, \beta_1, \beta_2, \beta_{12})$ and σ^2 (provide your chosen values), and $r \in \{1, 2, 5\}$. For each r :

- Generate 5,000 datasets from Design A and from Design B.
- Fit OLS including the interaction.

(a). Report **empirical bias** and **variance** for each coefficient under each design.

(b). Explain why bias is (approximately) zero for both designs and why variance is smaller under the factorial design.

Question 2. (20 points)

8.11 An article in *Industrial and Engineering Chemistry* ("More on Planning Experiments to Increase Research Efficiency," 1970, pp. 60–65) uses a 2^{5-2} design to investigate the effect of A = condensation temperature, B = amount of material 1, C = solvent volume, D = condensation time, and E = amount of material 2 on yield. The results obtained are as follows:

$$\begin{array}{llll} e = 23.2 & ad = 16.9 & cd = 23.8 & bde = 16.8 \\ ab = 15.5 & bc = 16.2 & ace = 23.4 & abcde = 18.1 \end{array}$$

- Verify that the design generators used were $I = ACE$ and $I = BDE$.
- Write down the complete defining relation and the aliases for this design.
- Estimate the main effects.

Question 3. (30 points)

8.31 Consider the following experiment:

Run	Treatment Combination
1	<i>d</i>
2	<i>ae</i>
3	<i>b</i>
4	<i>abde</i>
5	<i>cde</i>
6	<i>ac</i>
7	<i>bce</i>
8	<i>abcd</i>

Answer the following questions about this experiment:

- How many factors does this experiment investigate?
- How many independent factors are in this design?
- What are the generators for the dependent factors?
- Is this design a principal fraction?
- What is the complete defining relation and aliases?
- What is the resolution?

Question 4. (20 points)

Construct a 2^{7-2} design by choosing two four-factor interactions as the independent generators. Write down the complete alias structure for this design. What is the resolution of this design?